



CornellEngineering



Causal AI & Sustainability

Prof. Marcos López de Prado

Cornell University – College of Engineering

Lawrence Berkeley National Laboratory – U.S. Office of Science

ADIA Lab – UAE

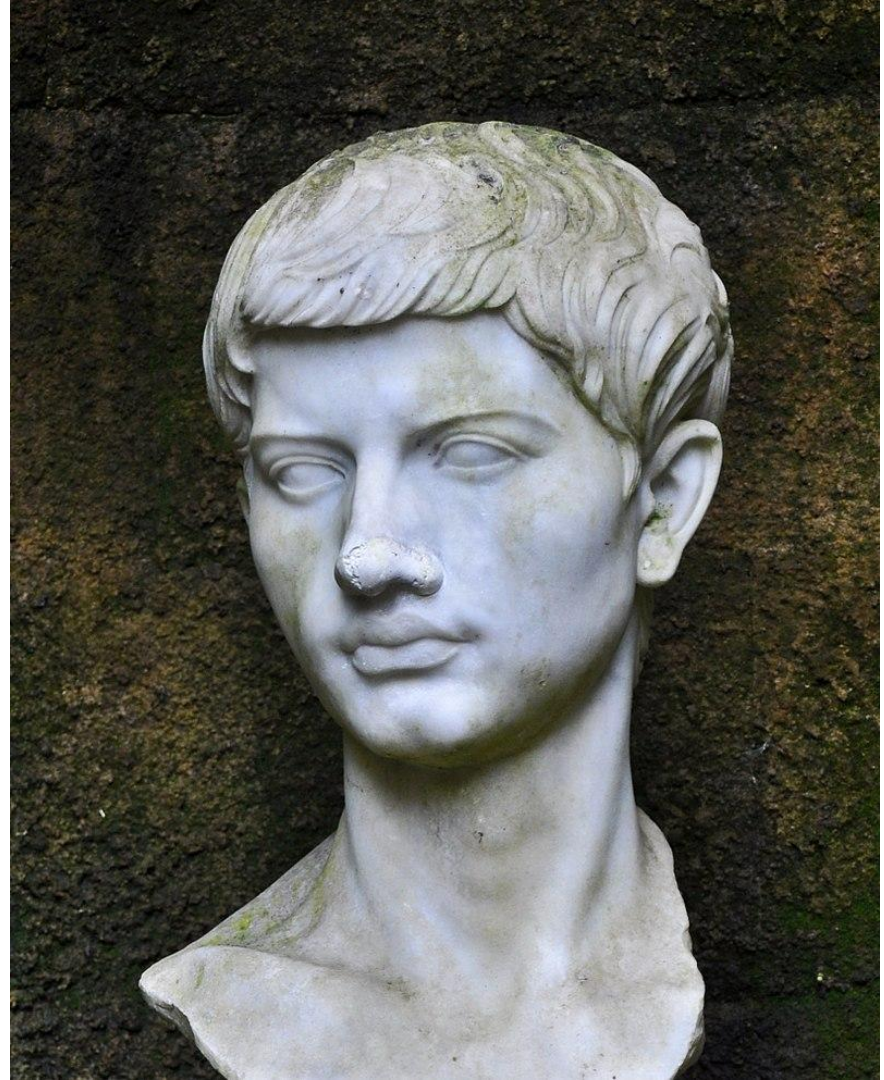
Seminar's Objective

- Every student of statistics learns that correlation does not imply causation
 - Association is an observational property
 - Causation is an interventional concept
- Causality plays a fundamental role in the scientific method
 - Scientific theories are falsifiable statements of the form “ X causes Y through mechanism M ”
- Much of modern scientific research is false due to the “replication crisis”:
 - **Type-A Spuriousity: Statistical flukes**
 - **Type-B Spuriousity: Non-causal association**
- **AI can optimize sustainability interventions only if it understands causation**, not just association
 - E.g., renewable-energy forecasting, emissions abatement, biodiversity preservation

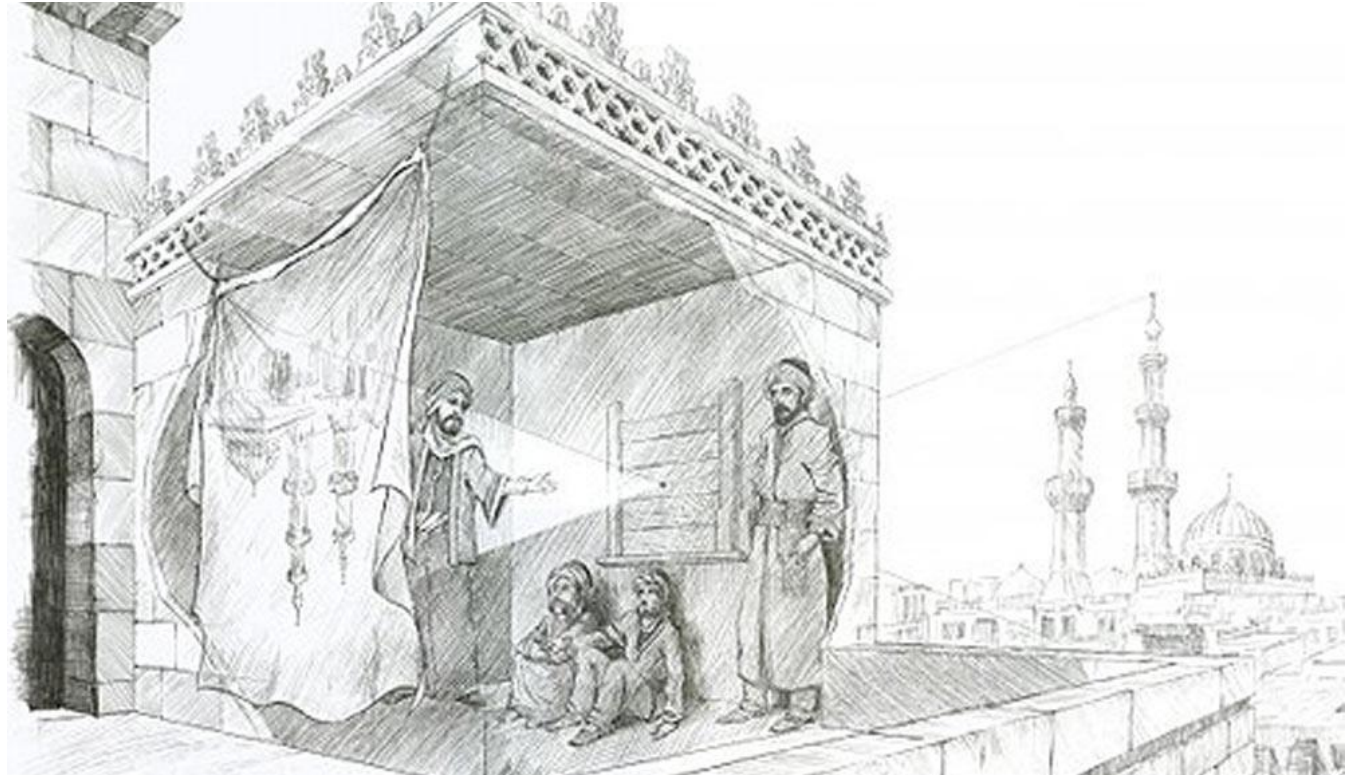
Why Study Cause and Effect?

*“Happy the man, who,
studying Nature’s laws,
thro’ known effects can
trace the secret cause”*

The Second Book of the Georgics
Publius Vergilius Maro, “Virgil” (70 – 19 BC)



Around the year 1011 that Arab mathematician Hasan Ibn Al Haytham (965 - 1040) proposed a scientific method for deducing causal mechanisms.



David Hume defined a cause-effect (causal) relation as that “where, if the first object had not been, the second never had existed.”

An Enquiry concerning Human
Understanding. Sec. VII. (1748)
David Hume (1711 - 1776)



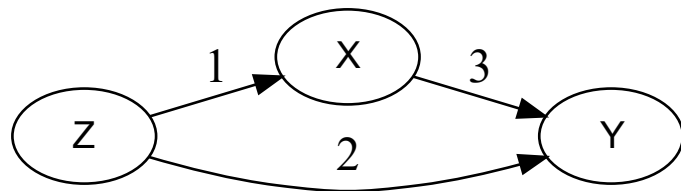
The Three Stages of the Scientific Method

Stage	Statement	Example
Phenomenological (induction)	X is associated with Y	CO ₂ levels are associated with global temperature
Theoretical (abduction)	X causes Y through mechanism M	CO ₂ causes warming via radiative forcing
Falsification Refutation (deduction)	Refutation attempts: • X does not cause Y • X causes Y, but not through mechanism M	Test through volcanic-eruption or solar-irradiance natural experiments

Association vs. Causation

Paths

- A **data-generating process** can be represented as a directed acyclic graph (DAG)
 - Nodes are **variables**
 - Arrows indicate the **direction of dependence**
- A **path** is a sequence of arrows and nodes that connect two variables X and Y , regardless of the direction of causation
- A **directed path** is a path where all arrows point in the same direction
- In a directed path that starts in X and ends in Y
 - X is an **ancestor** of Y , and
 - Y is a **descendant** of X



A DAG is a directed graph with no directed cycles.

In the DAG above, there are two paths between X and Y :

- a) A directed path: $X \rightarrow Y$
- b) A non-directed path: $X \leftarrow Z \rightarrow Y$

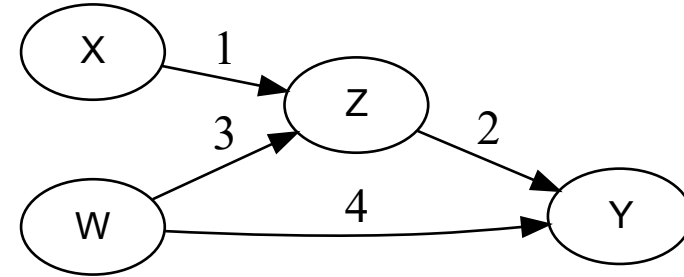
In the DAG above, X is a descendant of Z , and Y a descendant of X and Z .

Blocked Paths

- In a DAG with three variables $\{X, Y, Z\}$, a variable Z is
 - a **confounder** when the causal relationships include a structure $X \leftarrow Z \rightarrow Y$
 - a **collider** when the causal relationships are reversed, i.e. $X \rightarrow Z \leftarrow Y$
 - a **mediator** when the causal relationships include a structure $X \rightarrow Z \rightarrow Y$

A path between X and Y is **blocked** if either:

- a) the path traverses a collider, and the researcher has not conditioned on that collider or its descendants; or
- b) the researcher conditions on a variable in the path between X and Y , where the conditioned variable is not a collider



In the above DAG:

- W is a confounder to Z and Y
- Z is a collider to X and W
- Z is a mediator between X and Y

The path $X \rightarrow Z \leftarrow W \rightarrow Y$ is blocked by Z .

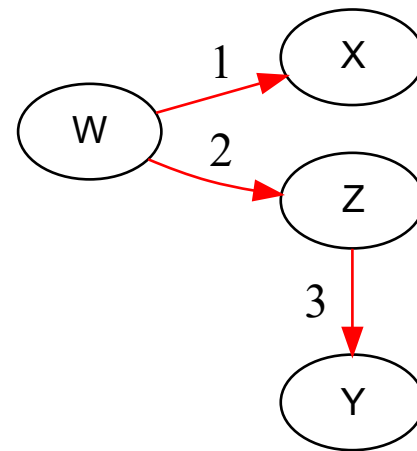
The only unblocked path between X and Y is the causal path, $X \rightarrow Z \rightarrow Y$.

What is Association?

- **Association** flows along an unblocked path between X and Y
 - Association is *symmetric* because paths do not follow the direction of causation
- Probabilistically, two variables X and Y are associated when knowing the value of one conveys information about the value of the other

$$\exists x, y | P[Y = y | X = x] \neq P[Y = y]$$

- **Statistical association is merely an observational statement on the joint distribution of probability**
 - $P[Y = y | X = x]$ does not measure the *effect* of X on Y

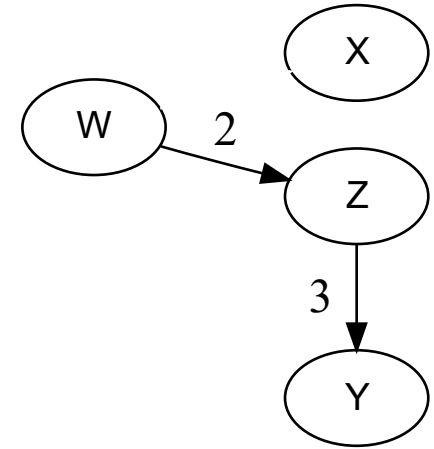


Weather (W) influences ice cream sales (X) and the number of swimmers (Z), hence the number of drownings (Y).

There is no directed path between X and Y , however X and Y are associated (red-dashed undirected edge), because of the unblocked path $X \leftarrow W \rightarrow Z \rightarrow Y$.

What is Causation?

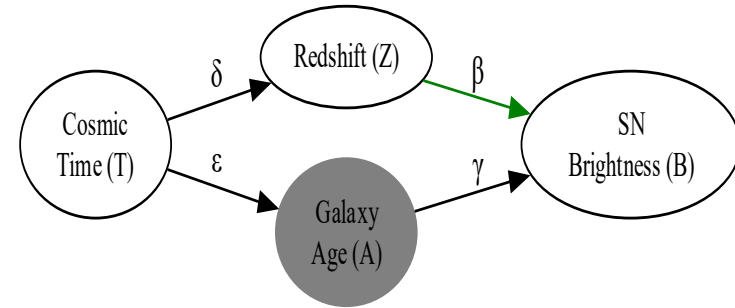
- **Causal association** flows along an unblocked *directed* path that starts in treatment X and ends in outcome Y , denoted the causal path
- Let $do[X = x]$ represent the do-operator on X
 - This is an intervention that sets the value of X to x , hence X is not influenced by any other variable
- **Definition:** X causes Y iff $P[Y|do[X]] > P[Y]$
- Association implies causation only if all non-causal paths are blocked
- Causality is
 - an interventional (beyond observational) concept
 - asymmetric (directional)
 - sequential: X happens first, and then Y adapts



A do-operation on X removes arrow (1), because X is no longer a function of W , while keeping all other things equal (“ceteris paribus”). As a result, there is no unblocked path between X and Y , and $P[Y|do[X]] = P[Y]$.

A Physics Example: Do We Need “Dark Energy”?

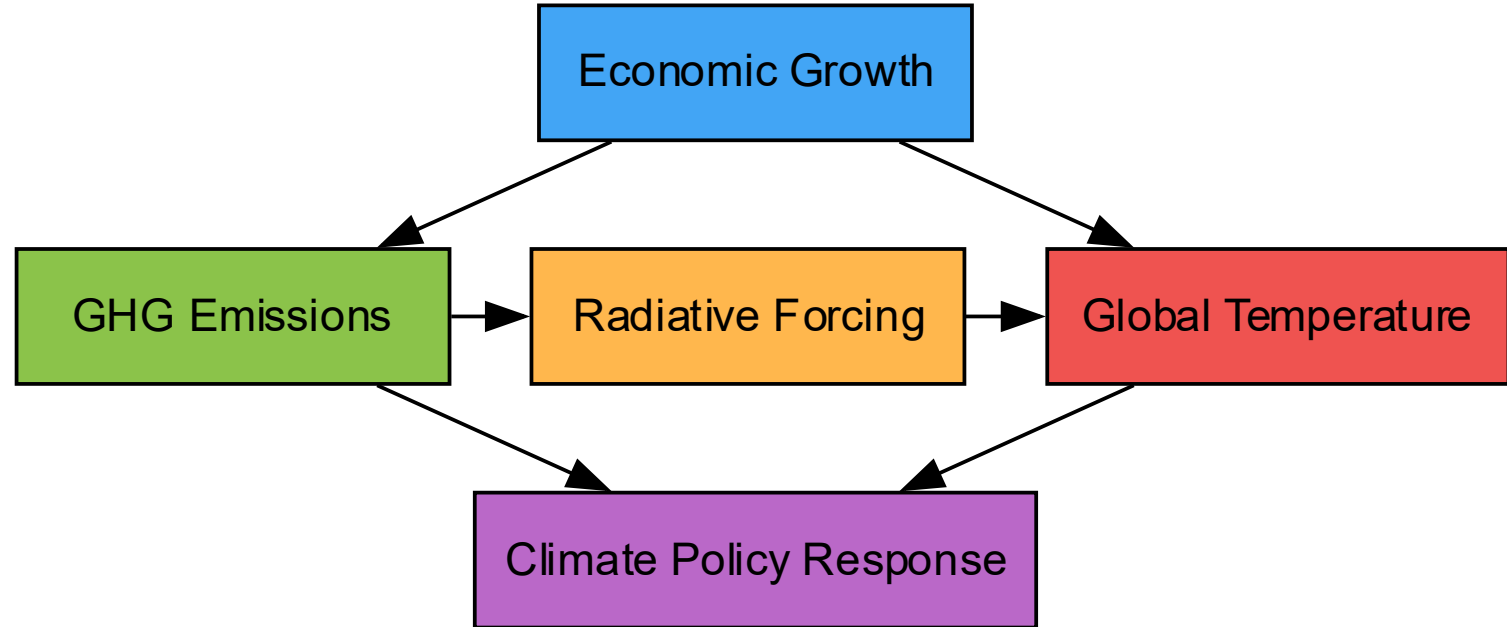
- The Supernova Cosmology Project (SCP) was established to measure the rate of expansion of the Universe, by comparing
 - the observed brightness of *Type Ia Supernovae (SNe)* with
 - the expected brightness given the observed redshift
- In 1998, the SCP found that SNe were dimmer than expected for a constant rate of expansion
 - SCP conjectured that “Dark Energy” may be responsible for the Universe’s accelerating expansion
- In 2025, [researchers](#) found that galaxy age is associated with both, brightness and redshift
 - This backdoor appears to explain the 1998 discrepancy
 - **Confounder bias: IF confirmed, there is no need for “Dark Energy”**



In the above causal graph, not controlling for the Galaxy_Age biases the Brightness prediction. An observed Brightness dimmer than predicted implied that the Universe’s expansion was accelerating, which led to the “Dark Energy” conjecture.

After controlling for Galaxy_Age, the SNe data no longer show acceleration and therefore do not require “Dark Energy.”

A Causal Graph for Climate Change



Blocking Non-Causal Paths

Interventional Studies

- In a **controlled experiment**, scientists assess causality by observing the effect on Y of changing the values of X , while keeping constant all other variables
 - E.g., [Ohm's law](#) of current, Newton's law of gravitation, etc.
- When some of the variables are not under direct experimental control, scientists may execute a **randomized controlled trial (RCT)**
 - E.g., Effectiveness of Pfizer's [COVID-19 vaccine](#)
- Under random assignment, subjects in the treatment group ($X = x_1$) are assumed to be indistinguishable from subjects in the control group ($X = x_0$)
 - Thanks to this assumption, the difference in outcomes can be attributed to the treatment

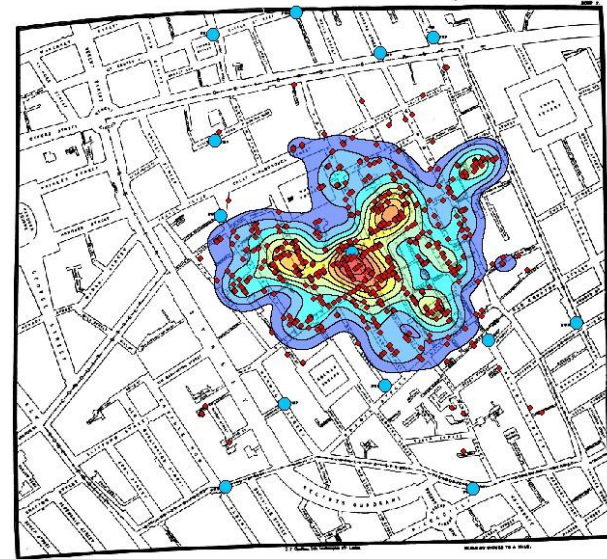


In the 1930s, Ronald Fisher popularized randomized experiments as a way to de-confound variables.

The first published RCT appeared in 1948. Today, well-blinded RCTs are considered the gold standard in experimental research.

Natural Experiments

- Sometimes interventional studies are not possible, because they are unfeasible, unethical, or prohibitively expensive
- In a natural experiment, subjects are assigned to the treatment and control groups determined randomly by Nature or by other factors not controlled by scientists
- Examples of natural experiments include
 - **Random shocks:** When an unexpected event affects one group (treatment) but not another (control)
 - **Regression discontinuity design (RDD):** When treatment and control groups are comparable in everything but the slight difference in the assignment variable, attributed to noise
 - **Difference-in-differences studies (DID):** When both groups are affected by a common trend over time, the effect is measured as change in treatment group minus the change in control group



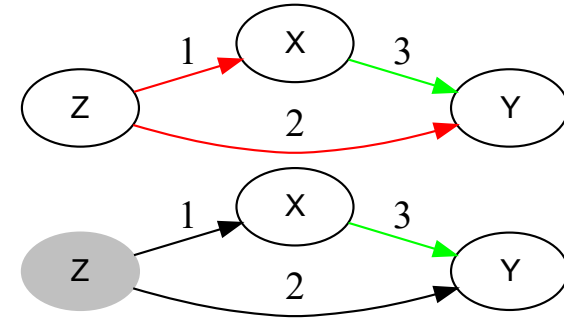
In 1854, Dr. John Snow found that exposure to contaminated water causes cholera. Sick and healthy neighbors of London's Soho district were comparable in all respects, except by their use of different water pumps.

Simulated Intervention: Backdoor Adjustment

- Under some conditions, we can simulate an intervention
- A **backdoor path** between X and Y is an unblocked non-causal path that connects those two variables
- A set of variables S satisfies the **backdoor criterion** if the following two conditions are true:
 - conditioning on S blocks all backdoor paths between X and Y
 - S does not contain any descendants of X
- Then, S is a sufficient adjustment set, and the causal effect of X on Y can be estimated as:

$$P[Y = y | \text{do}[X = x]] = \sum_s P[Y = y | X = x, S = s] P[S = s]$$

- Examples of other adjustments: Front-door, IV, etc.



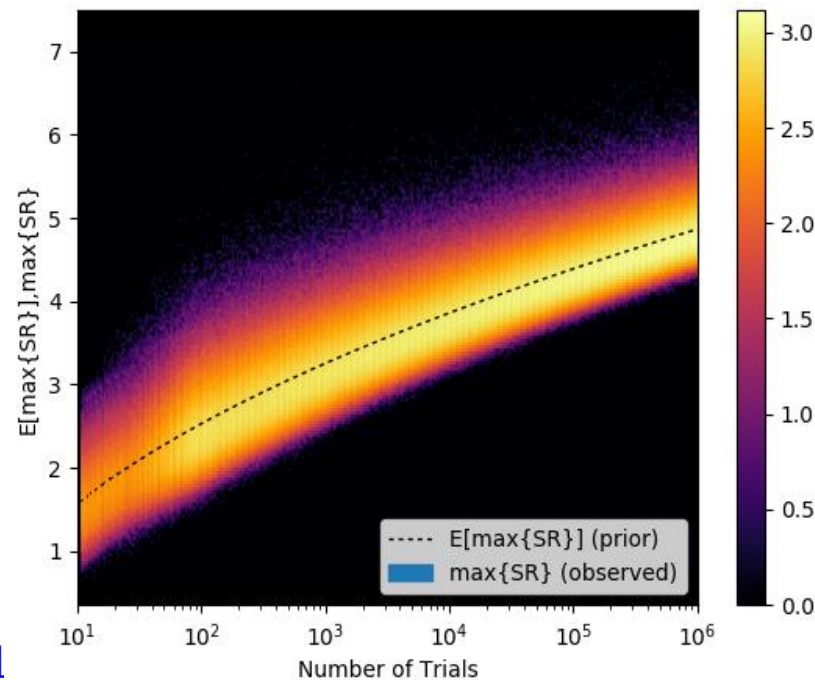
Conditioning on confounder Z (shaded node) blocks the path $X \leftarrow Z \rightarrow Y$, leaving the causal path $X \rightarrow Y$ as the only unblocked path.

Under those circumstances, association does imply causation, and we can simulate the outcome of a do-operation through conditional probabilities.

Spurious Results

Type-A Spuriousity

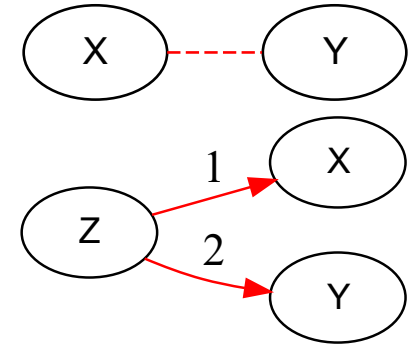
- Type-A spuriousity occurs when a researcher mistakes random variability (noise) for signal, resulting in a *false association*
- Type-A spuriousity has several attributes:
 - a) it results in type-1 errors (false positives)
 - b) for the same number of trials, it has a lower probability to take place as the sample size grows
 - c) it can be corrected through multiple-testing adjustments
- Two main reasons for Type-A spuriousity
 - *p*-hacking, e.g., [Hochberg \[1988\]](#)
 - Backtest overfitting, [Bailey and López de Prado \[2014\]](#)
- “Why most published research findings are false” (John Ioannidis [2005])



Distribution of the maximum t-value as a function of the number of trials, where the true t-value is zero. See “[The False Strategy Theorem](#).”

Type-B Spuriousity

- Type-B spuriousity occurs when a researcher mistakes association for causation (e.g., due to misspecification)
- Type-B spuriousity has several attributes:
 - a) it results in type-1 errors **and type-2 errors** (false positives and false negatives);
 - b) it can occur **with a single trial**;
 - c) it has a **greater probability to take place as the sample size grows**, because the non-causal association can be estimated with lower error; and
 - d) it **cannot be corrected through multiple-testing adjustments**. Its correction requires the injection of extra-statistical information, in the form of a causal theory
- Type-B spurious factors exhibit
 - misattributed causes, due to confounder bias, collider bias, etc.
 - time-varying causes, under non-causal parameter shift

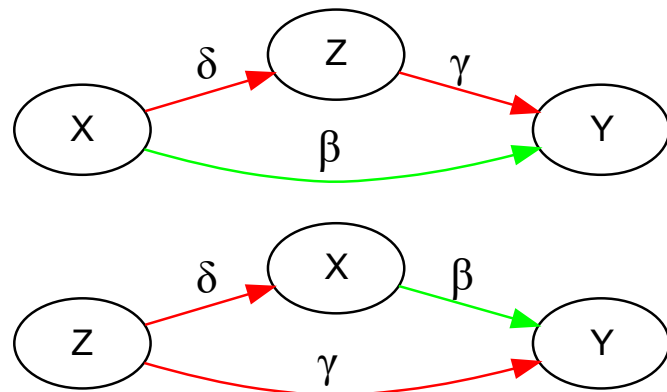


The top graph is an example of false association (type-A spuriousity). The bottom graph is an example of association mistaken for causation (type-B spuriousity).

Type-A and type-B spuriousity are mutually exclusive. For type-B spuriousity to take place, the association must be non-causal but true, which precludes that association from being type-A spurious.

Type-B(1) Spuriousity: Under-Controlling

- Consider a researcher who fits $Y = X\beta + \varepsilon$ on data generated by $Y := X\beta + Z\gamma + u$, where $\gamma \neq 0$ and u is white noise
 - As a consequence, $E[\varepsilon|X] = \gamma E[Z|X]$
 - $E[Z|X] \neq 0 \Rightarrow E[\varepsilon|X] \neq 0$ (exogeneity is not satisfied)
- Case 1: Z is a mediator ($Z := X\delta + v$, with $\delta \neq 0$)
 - the chosen specification biases $\hat{\beta}$
 - however $\hat{\beta}$ can still be interpreted as a total causal effect
- Case 2: Z is a confounder ($X := Z\delta + v$, with $\delta \neq 0$)
 - the chosen specification biases $\hat{\beta}$
 - $\hat{\beta}$ cannot be interpreted as a causal effect (direct or total)

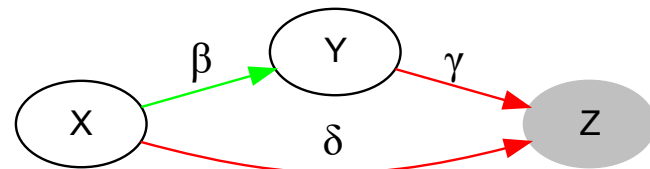
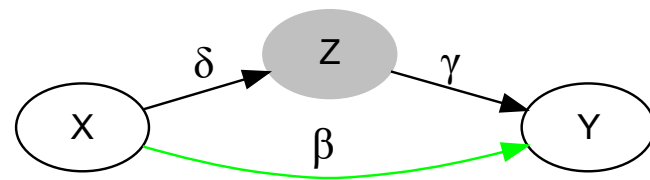


Statistics textbooks often treat all missing variables as equal. **This is a mistake.**

In the top graph, Z is a mediator, and missing Z has mild consequences. In the bottom graph, Z is a confounder, and missing Z will likely lead to false positives or false negatives.

Type-B(2) Spuriousity: Over-Controlling

- Statisticians have been trained for decades to control for any variable Z associated with Y that is not X
 - Statistics textbooks dismiss as a harmless error the inclusion of an irrelevant variable, regardless of the variable's role in the causal graph
- Case 1: Z is a mediator
 - Controlling for a mediator interferes with the mediated effect and the total effect, which the researcher may wish to assess
 - $\hat{\beta}$ measures only the direct effect
- Case 2: Z is a collider
 - Controlling for a collider opens a backdoor path, $X \rightarrow Z \leftarrow Y$ (Berkson's fallacy)

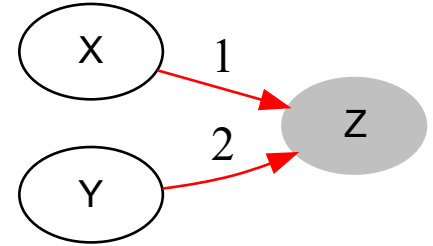


Greene [2012, section 4.3.3] states that the only downside to adding superfluous variables is a reduction in the precision of the estimates. **This is a mistake.**

Over-controlling for a collider has the same consequences as under-controlling for a confounder: **an open backdoor.**

Type-B(3) Spuriousity: Specification-Searching

- The use of explanatory power (an associational, non-causal concept) for selecting the specification of a factor model is inconsistent with that model's causal content
- Specification-searching commingles two separate and sequential stages of the causal analysis:
 - 1) **Causal discovery:** Finding the causal graph
 - 2) **Control:** Use the graph to determine the correct specification
- Stage (2) should be informed by stage (1), not the other way around
 - A researcher may achieve higher explanatory power by combining multiple causes of Y , at the expense of biasing the multiple parameters' estimates due to multicollinearity or over-controlling for a collider



Empirical studies often justify the chosen specification in terms of explanatory power. This commingles causal discovery with controlling, and **all but ensures that the regressors will include colliders** (Berkson's fallacy).

Monte Carlo Experiments

Type-B(1) Spuriousity : Forks

Confounders

- Consider the fork structure in the right graph
- Applying Bayesian network factorization

$$P[X, Y, Z] = P[Z]P[X|Z]P[Y|Z]$$

- X and Y are associated, since

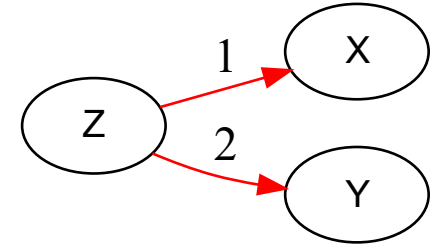
$$P[X, Y] = \sum_Z P[Z]P[X|Z]P[Y|Z] \neq P[X]P[Y]$$

- This is an example of non-causal association

- X and Y are associated through the backdoor path

$$Y \leftarrow Z \rightarrow X$$

- Given the causal content of the factor model, a statistically significant $\hat{\beta}$ implies that X causes Y
 - This claim of statistical significance is type-B spurious



$$Y_t = \alpha + \beta X_t + \varepsilon_t$$

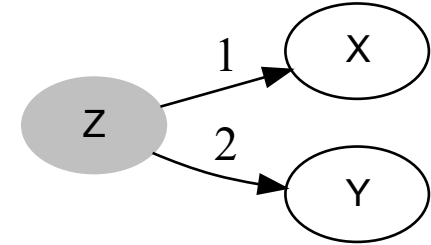
OLS Regression Results						
=====						
Dep. Variable:	Y		R-squared:	0.247		
Model:	OLS		Adj. R-squared:	0.247		
Method:	Least Squares		F-statistic:	1640.		
Date:	Sun, 14 Aug 2022		Prob (F-statistic):	2.69e-310		
Time:	13:14:32		Log-Likelihood:	-8052.6		
No. Observations:	5000		AIC:	1.611e+04		
Df Residuals:	4998		BIC:	1.612e+04		
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0090	0.017	0.524	0.600	-0.025	0.043
X	0.4964	0.012	40.493	0.000	0.472	0.520
=====						
Omnibus:	1.784		Durbin-Watson:		1.964	
Prob(Omnibus):	0.410		Jarque-Bera (JB):		1.746	
Skew:	0.027		Prob(JB):		0.418	
Kurtosis:	3.073		Cond. No.		1.40	
=====						

The Backdoor Adjustment

- The effect of conditioning by Z is equivalent to simulating a do-operation (an intervention)
 - It blocks the backdoor path, resulting in the conditional independence of X and Y ,

$$P[X, Y|Z] = \frac{P[X, Y, Z]}{P[Z]} = P[X|Z]P[Y|Z]$$



$$Y_t = \alpha + \beta X_t + \gamma Z_t + \varepsilon_t$$

- It is possible to remove the confounder-induced bias by adding Z as a regressor (the **partial correlations method**)
- With the correct model specification, the researcher concludes that X does not cause Y

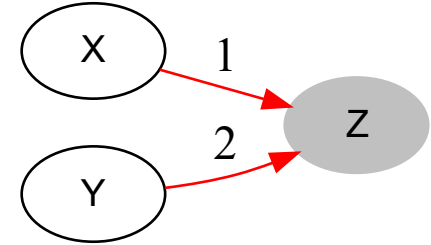
OLS Regression Results						
=====						
Dep. Variable:	Y	R-squared:	0.495			
Model:	OLS	Adj. R-squared:	0.495			
Method:	Least Squares	F-statistic:	2447.			
Date:	Sun, 14 Aug 2022	Prob (F-statistic):	0.00			
Time:	13:14:32	Log-Likelihood:	-7054.9			
No. Observations:	5000	AIC:	1.412e+04			
Df Residuals:	4997	BIC:	1.414e+04			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0054	0.014	0.383	0.702	-0.022	0.033
X	0.0007	0.014	0.051	0.959	-0.027	0.029
Z	0.9957	0.020	49.506	0.000	0.956	1.035
=====						
Omnibus:	2.685	Durbin-Watson:	1.972			
Prob(Omnibus):	0.261	Jarque-Bera (JB):	2.629			
Skew:	0.050	Prob(JB):	0.269			
Kurtosis:	3.050	Cond. No.	2.62			
=====						

Type-B(2) Spuriousity : Immoralities

Colliders

- This causal graph shows a collider:
 - Variable Z is influenced by both, the treatment X and the outcome Y
- If a researcher controls for Z , the result is **a false positive** (bottom table)
- Compare the fork structure with the immorality structure
 - When the direction of causality is reversed, a confounder becomes a collider
 - **The direction of causality is critical for specification**
- One problem is, the direction of causality cannot always be determined from data
 - Causal graphs incorporates extra-statistical (beyond observational) information



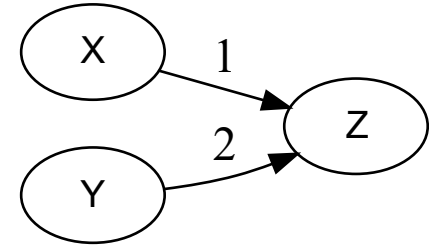
$$Y_t = \alpha + \beta X_t + \gamma Z_t + \varepsilon_t$$

OLS Regression Results						
=====						
Dep. Variable:	Y	R-squared:	0.499			
Model:	OLS	Adj. R-squared:	0.499			
Method:	Least Squares	F-statistic:	2490.			
Date:	Sun, 14 Aug 2022	Prob (F-statistic):	0.00			
Time:	13:11:51	Log-Likelihood:	-5314.4			
No. Observations:	5000	AIC:	1.063e+04			
Df Residuals:	4997	BIC:	1.065e+04			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0138	0.010	-1.388	0.165	-0.033	0.006
X	-0.4963	0.012	-40.405	0.000	-0.520	-0.472
Z	0.4988	0.007	70.575	0.000	0.485	0.513
=====						
Omnibus:	0.058	Durbin-Watson:	1.998			
Prob(Omnibus):	0.971	Jarque-Bera (JB):	0.037			
Skew:	0.001	Prob(JB):	0.982			
Kurtosis:	3.013	Cond. No.	2.41			

Berkson's Fallacy

- [Berkson's fallacy](#) occurs when a spurious association is observed between two independent variables, as a result of conditioning on a collider
- With a careful selection of colliders, a researcher can present evidence in support of any spurious investment factor
- The correct causal treatment of a collider is to indicate its presence, and justify why researchers should not control for it
- Over-controlling leads to
 - false positives, in the presence of colliders
 - false negatives, in the presence of mediators
 - E.g., controlling for Z in $X \rightarrow Z \rightarrow Y$



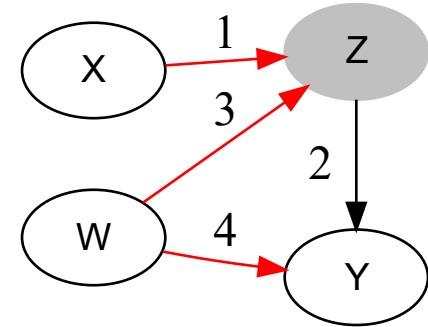
$$Y_t = \alpha + \beta X_t + \varepsilon_t$$

OLS Regression Results						
Dep. Variable:	Y	R-squared:	0.000			
Model:	OLS	Adj. R-squared:	-0.000			
Method:	Least Squares	F-statistic:	0.01120			
Date:	Sun, 14 Aug 2022	Prob (F-statistic):	0.916			
Time:	13:11:51	Log-Likelihood:	-7043.2			
No. Observations:	5000	AIC:	1.409e+04			
Df Residuals:	4998	BIC:	1.410e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-0.0221	0.014	-1.580	0.114	-0.050	0.005
X	0.0015	0.014	0.106	0.916	-0.026	0.029
Omnibus:	0.633	Durbin-Watson:	1.998			
Prob(Omnibus):	0.729	Jarque-Bera (JB):	0.638			
Skew:	0.028	Prob(JB):	0.727			
Kurtosis:	2.994	Cond. No.	1.02			

Type-B(3) Spuriousity : Chains

Confounded Mediators

- This causal graph shows a mediator and a confounder:
 - Variable Z mediates the causal flow from the treatment X to the outcome Y
 - Variable W confounds Z and Y
- If a researcher controls for Z , the outcome is a **false positive** (bottom table)
 - The reason is that Z also operates as a collider to X and W
 - Controlling for Z opens a backdoor path $X \rightarrow Z \leftarrow W \rightarrow Y$
 - While it is true that X causes Y (through Z), the collider's bias is so strong that the sign of the relationship is reversed ($\hat{\beta} \ll 0$)
- In the absence of link 3, controlling for Z would have led to a **false negative**

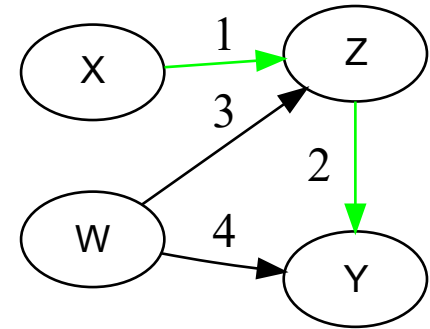


$$Y_t = \alpha + \beta X_t + \gamma Z_t + \varepsilon_t$$

OLS Regression Results						
Dep. Variable:	Y	R-squared:	0.784			
Model:	OLS	Adj. R-squared:	0.784			
Method:	Least Squares	F-statistic:	9069.			
Date:	Sun, 14 Aug 2022	Prob (F-statistic):	0.00			
Time:	13:04:29	Log-Likelihood:	-8061.9			
No. Observations:	5000	AIC:	1.613e+04			
Df Residuals:	4997	BIC:	1.615e+04			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0027	0.017	0.160	0.873	-0.031	0.036
X	-0.4814	0.021	-22.621	0.000	-0.523	-0.440
Z	1.4899	0.012	121.680	0.000	1.466	1.514
Omnibus:	0.314	Durbin-Watson:	1.994			
Prob(Omnibus):	0.855	Jarque-Bera (JB):	0.267			
Skew:	0.000	Prob(JB):	0.875			
Kurtosis:	3.036	Cond. No.	2.41			

Mediation Fallacy & Simpson's Paradox

- [The Mediation Fallacy](#) involves conditioning on the mediator when the mediator and the outcome are confounded
- [Simpson's paradox](#) occurs when there is an association in several groups, but it disappears or reverses when the groups are combined
- The solution to Simpson's paradox is to inject extra-statistical information in the form of a causal graph
- We can estimate the unbiased effect ($\hat{\beta} \gg 0$)
 - Specification-searching would have returned a misspecified model (R^2 drops from 0.78 to 0.14!)
 - Adding W increases R^2 to 0.71 (still below 0.78)



$$Y_t = \alpha + \beta X_t + \varepsilon_t$$

OLS Regression Results

Dep. Variable:	Y	R-squared:	0.144			
Model:	OLS	Adj. R-squared:	0.144			
Method:	Least Squares	F-statistic:	840.8			
Date:	Sun, 14 Aug 2022	Prob (F-statistic):	5.32e-171			
Time:	13:04:29	Log-Likelihood:	-11504.			
No. Observations:	5000	AIC:	2.301e+04			
Df Residuals:	4998	BIC:	2.303e+04			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0222	0.034	-0.650	0.515	-0.089	0.045
X	1.0055	0.035	28.996	0.000	0.938	1.073
=====						
Omnibus:	0.250	Durbin-Watson:	1.993			
Prob(Omnibus):	0.883	Jarque-Bera (JB):	0.288			
Skew:	0.009	Prob(JB):	0.866			
Kurtosis:	2.968	Cond. No.	1.02			

Causal Discovery

Causal Discovery Algorithms

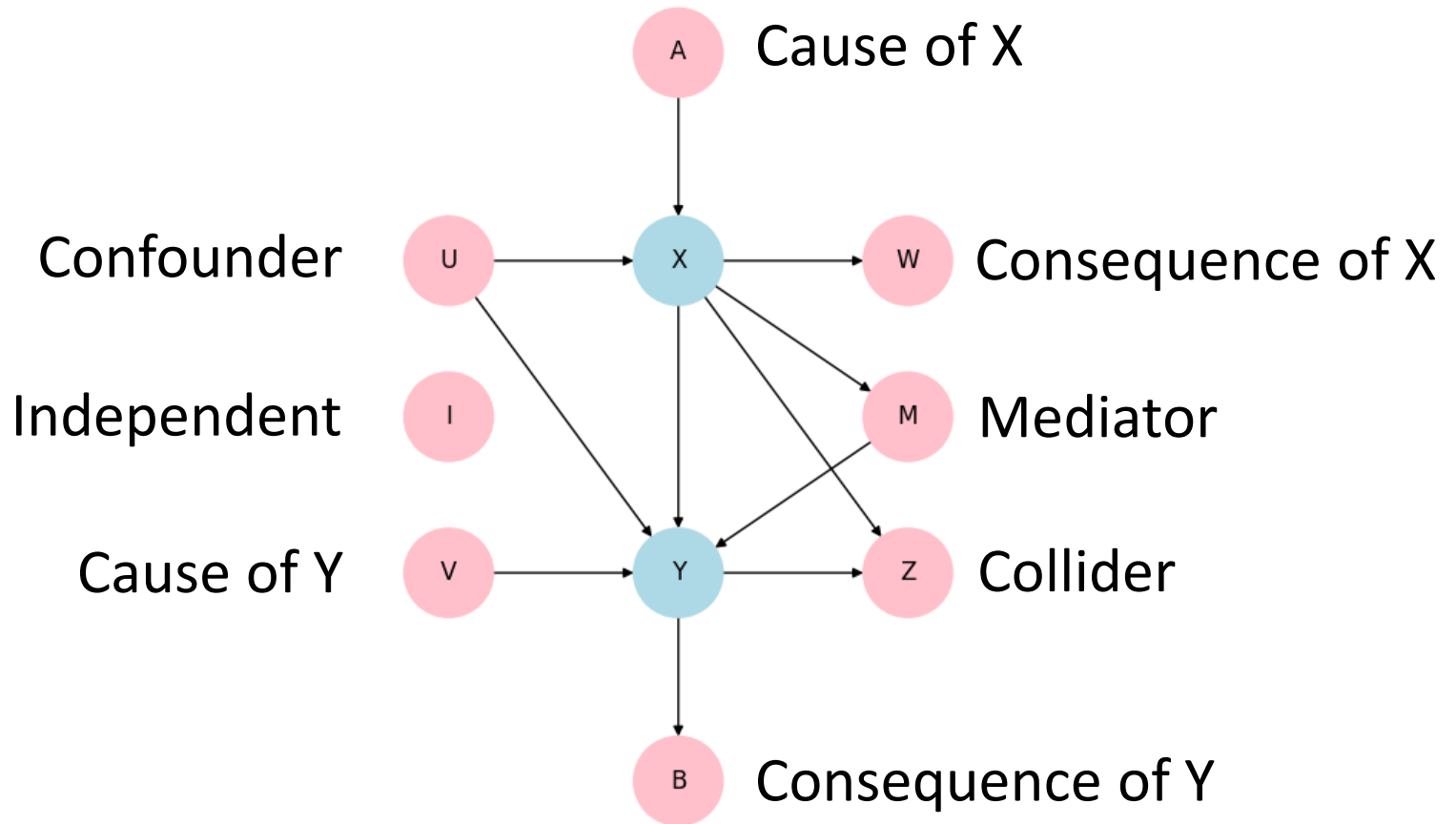
- Over the past three decades, statisticians have developed numerous computational methods and algorithms for the discovery of causal relations, represented as directed acyclic graphs (DAGs)
- These methods can be divided into three types:
 - **Score-based algorithms** optimize a defined score function, and can be used in the absence latent confounders (e.g., GES)
 - **Constraint-based methods** exploit conditional independence relationships in the data (e.g., PC, FCI)
 - FCI can give asymptotically correct results even if there are latent confounders
 - **Functional causal models** (FCMs) distinguish between DAGs in the same equivalence class



Abstract

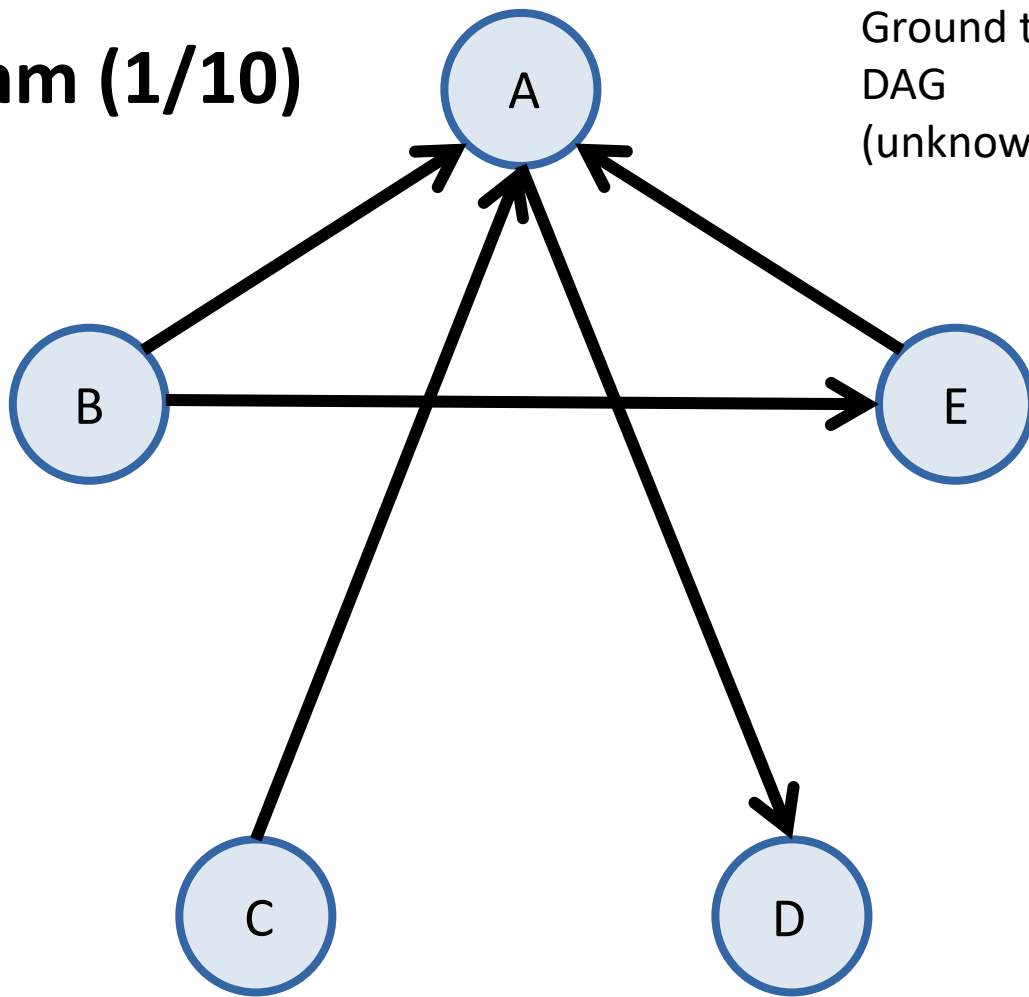
Previous asymptotically correct algorithms for recovering causal structure from sample probabilities have been limited even in sparse causal graphs to a few variables. We describe an asymptotically correct algorithm whose complexity for fixed graph connectivity increases polynomially in the number of vertices, and may in practice recover sparse graphs with several hundred variables. From sample data with $n = 20,000$, an implementation of the algorithm on a DECStation 3100 recovers the edges in a linear version of the ALARM network with 37 vertices and 46 edges. Fewer than 8% of the undirected edges are incorrectly identified in the output. Without prior ordering information, the program also determines the direction of edges for the ALARM graph with an error rate of 14%. Processing time is less than 10 seconds. *Keywords* DAGs, Causal Modelling.

Discovering a causal graph typically requires extra-statistical (beyond observational) information, in the form of an experiment or expert knowledge. However, under some assumptions, it is possible to derive causal graphs that are *consistent with* the observed sample.

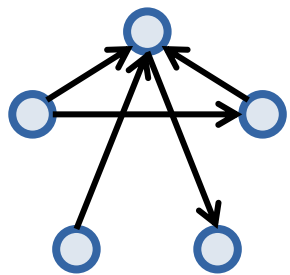


PC Algorithm (1/10)

Ground truth
DAG
(unknown)

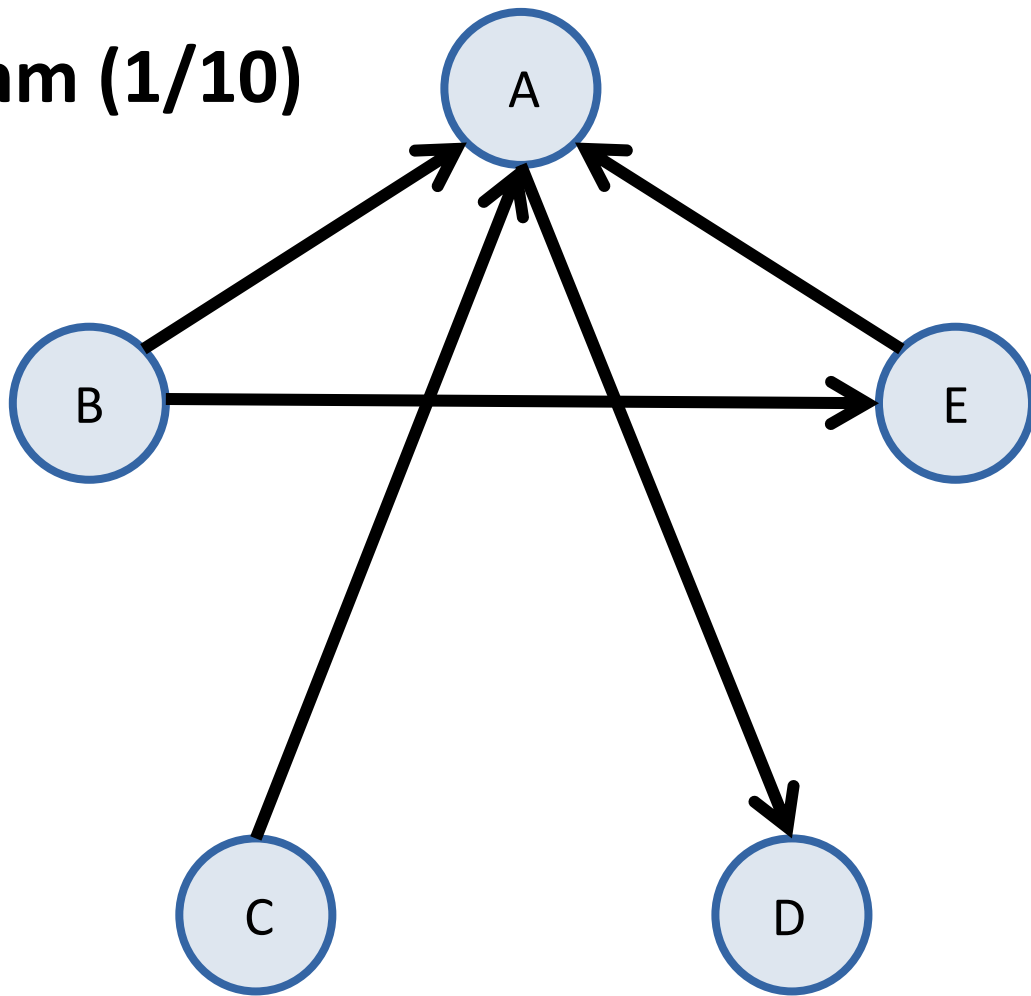


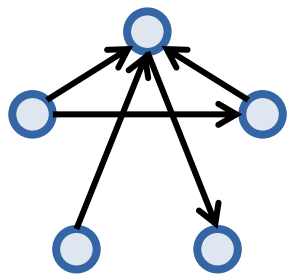
- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders



PC Algorithm (1/10)

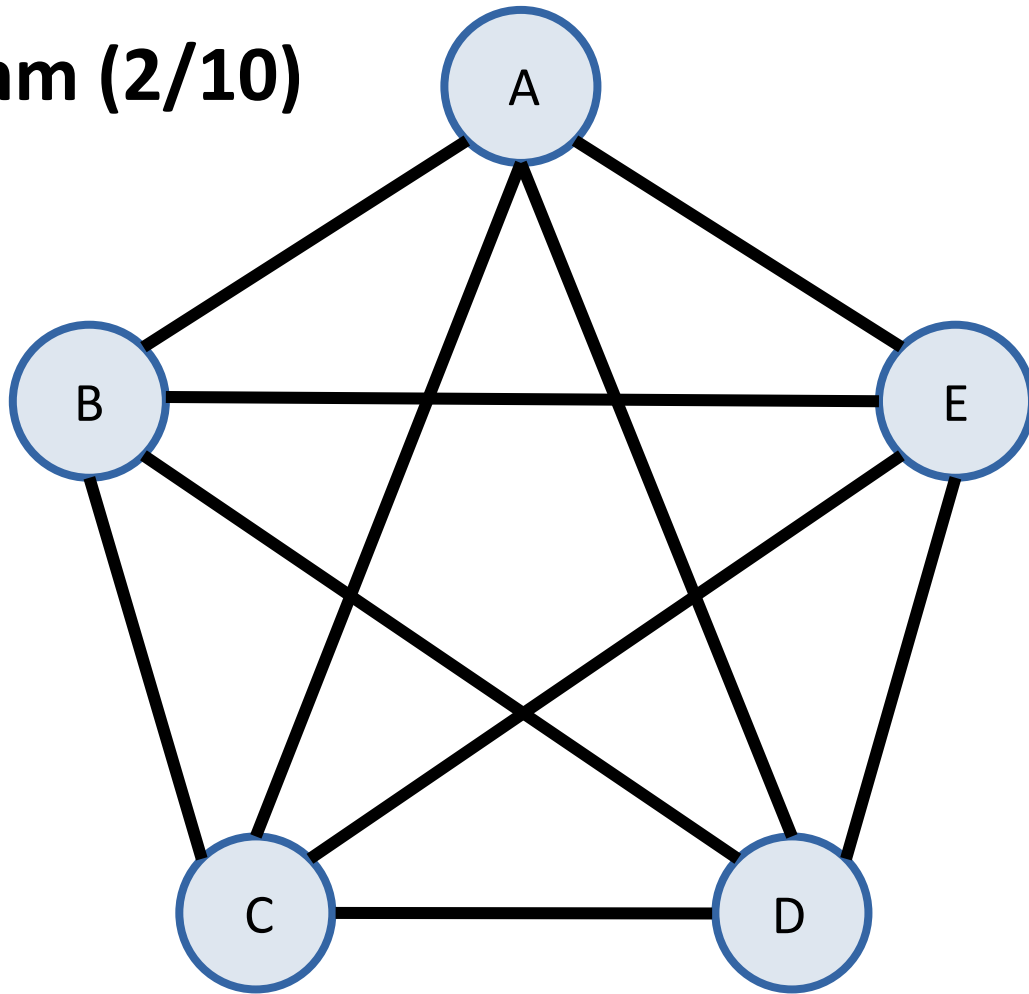
- Start with a complete (undirected) graph
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- Propagate the orientation, assuming we have found all the colliders

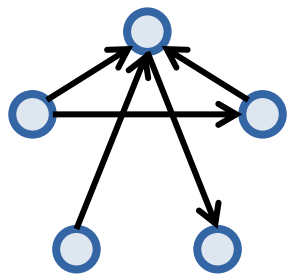




PC Algorithm (2/10)

- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders

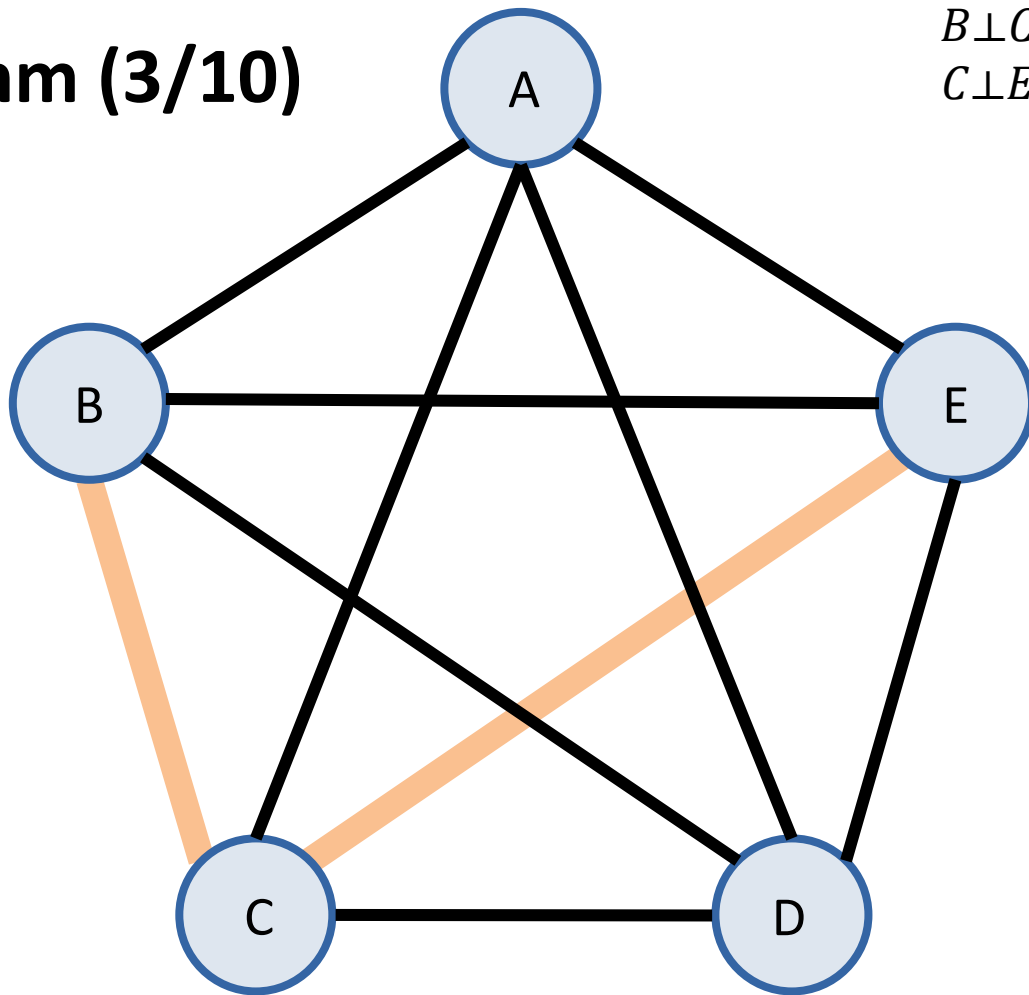


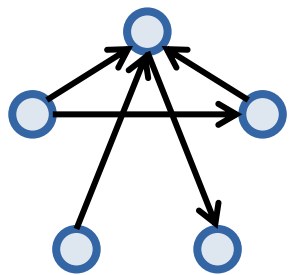


PC Algorithm (3/10)

$B \perp C$
 $C \perp E$

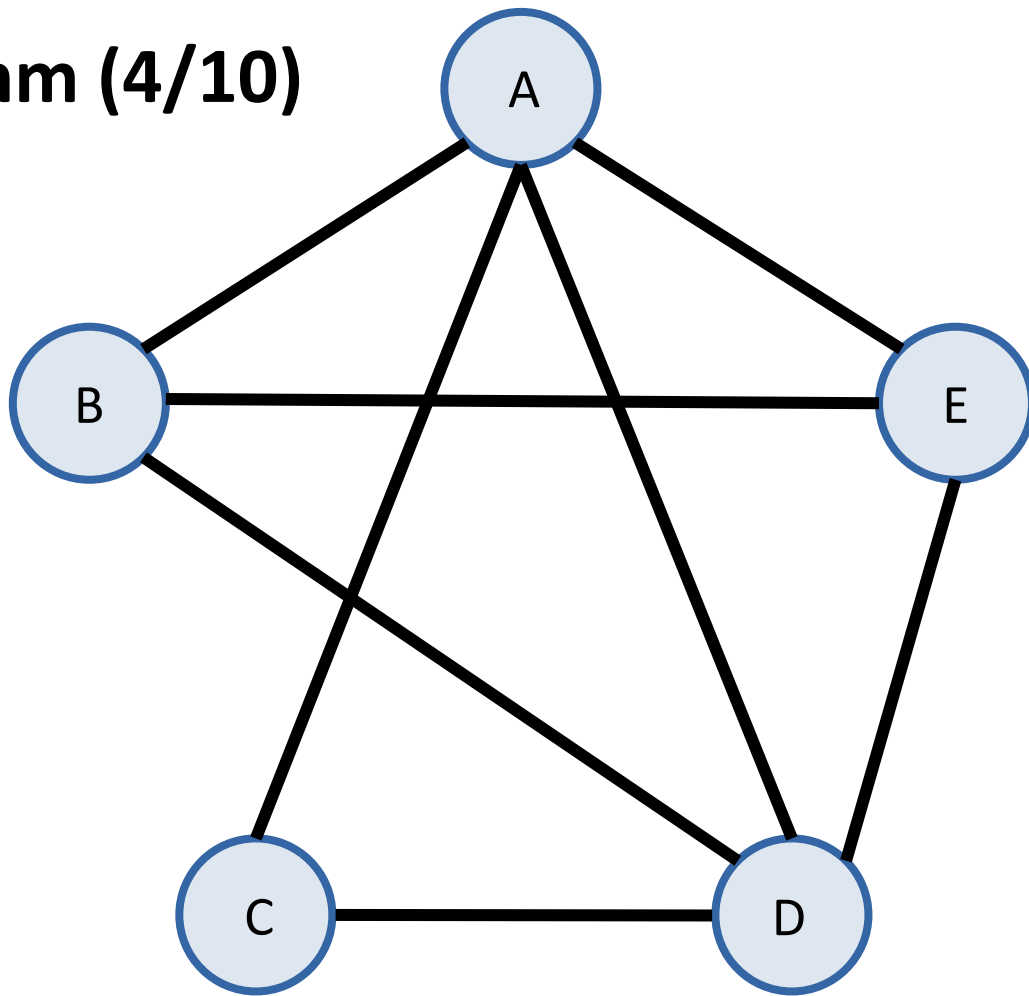
- Start with a complete (undirected) graph
- Remove the edge $X - Y$ if $X \perp Y | Z$ for some (possibly empty) set of nodes Z
- For all $X - Z - Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y) | Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders

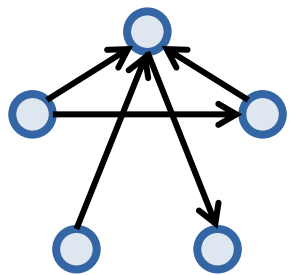




PC Algorithm (4/10)

- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders

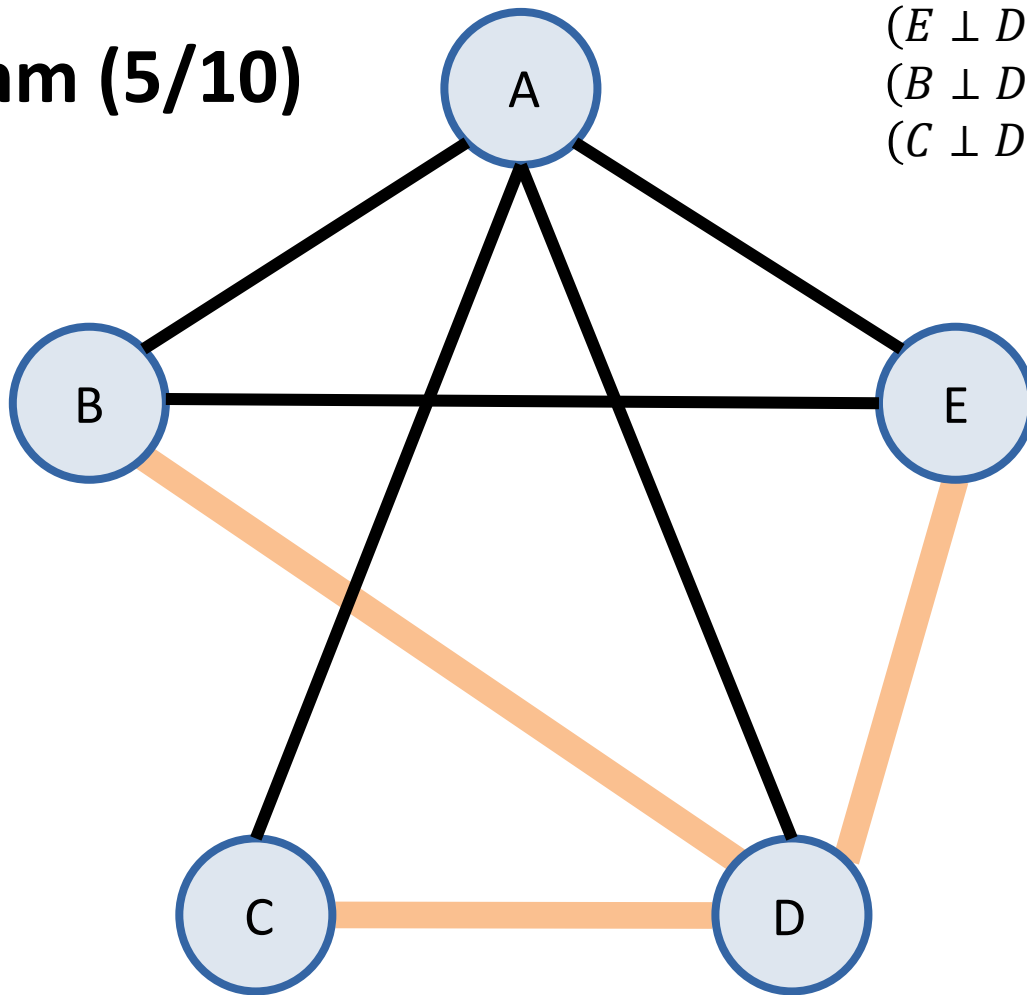


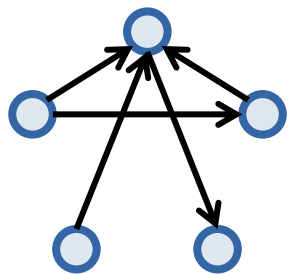


PC Algorithm (5/10)

$$\begin{aligned} &(E \perp D)|A \\ &(B \perp D)|A \\ &(C \perp D)|A \end{aligned}$$

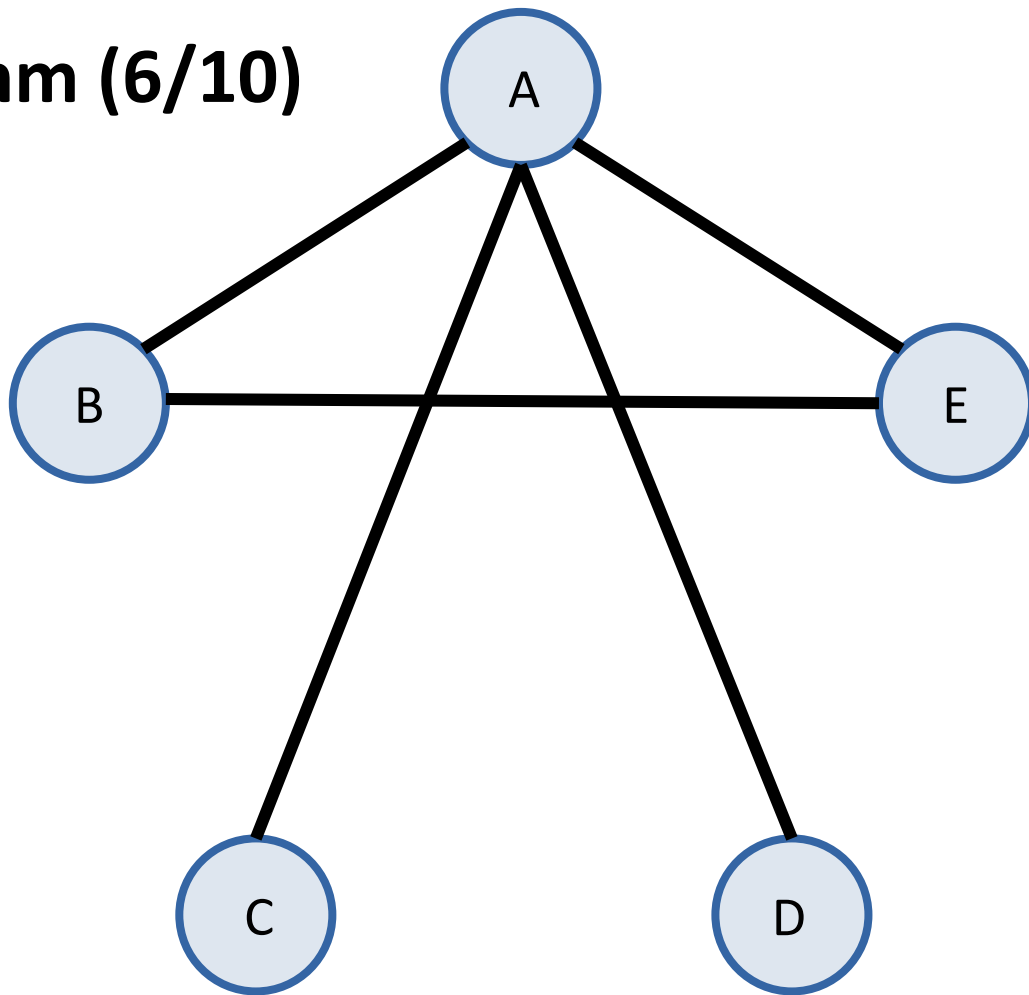
- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders

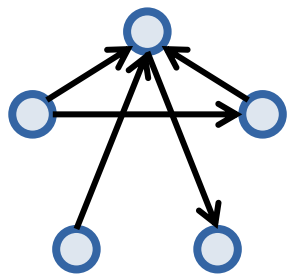




PC Algorithm (6/10)

- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders



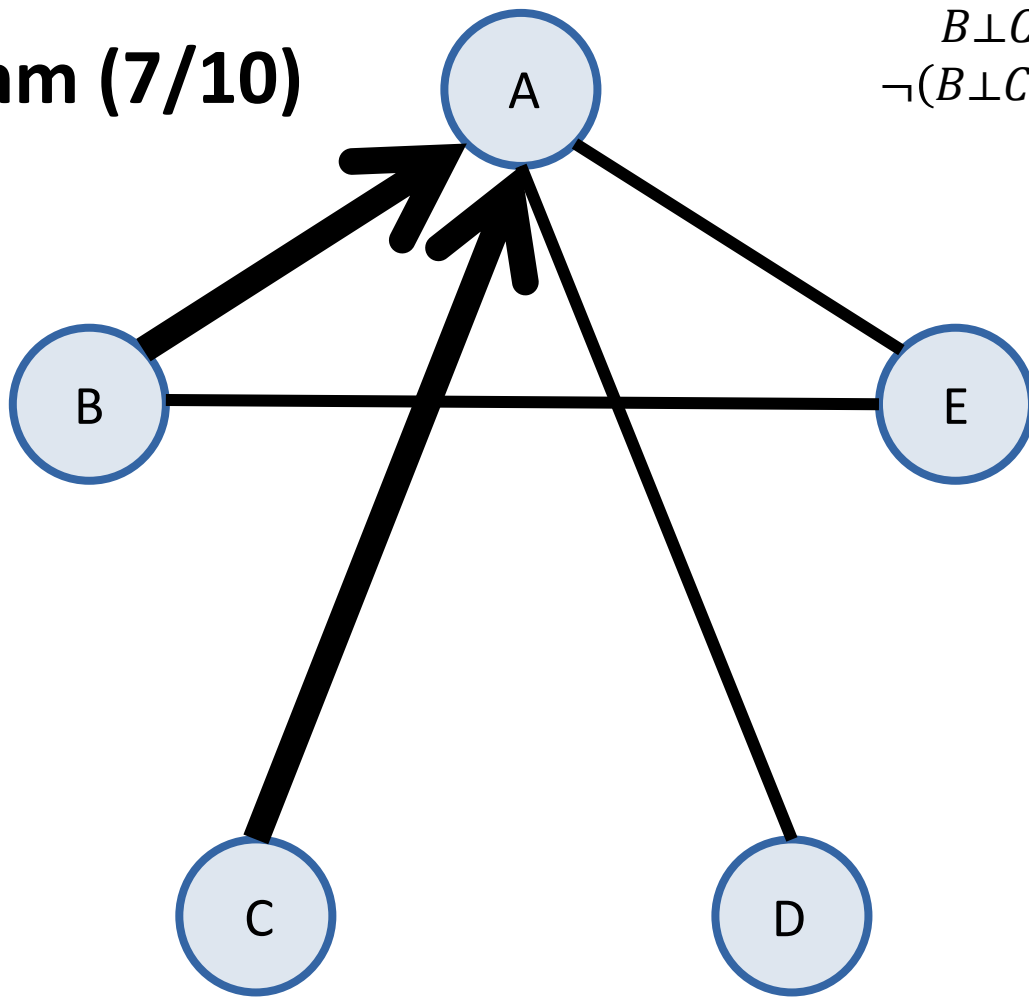


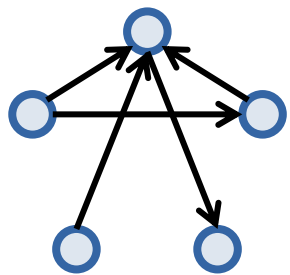
PC Algorithm (7/10)

$$B \perp C$$

$$\neg(B \perp C) | A$$

- Start with a complete (undirected) graph
- Remove the edge $X - Y$ if $X \perp Y | Z$ for some (possibly empty) set of nodes Z
- For all $X - Z - Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y) | Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders



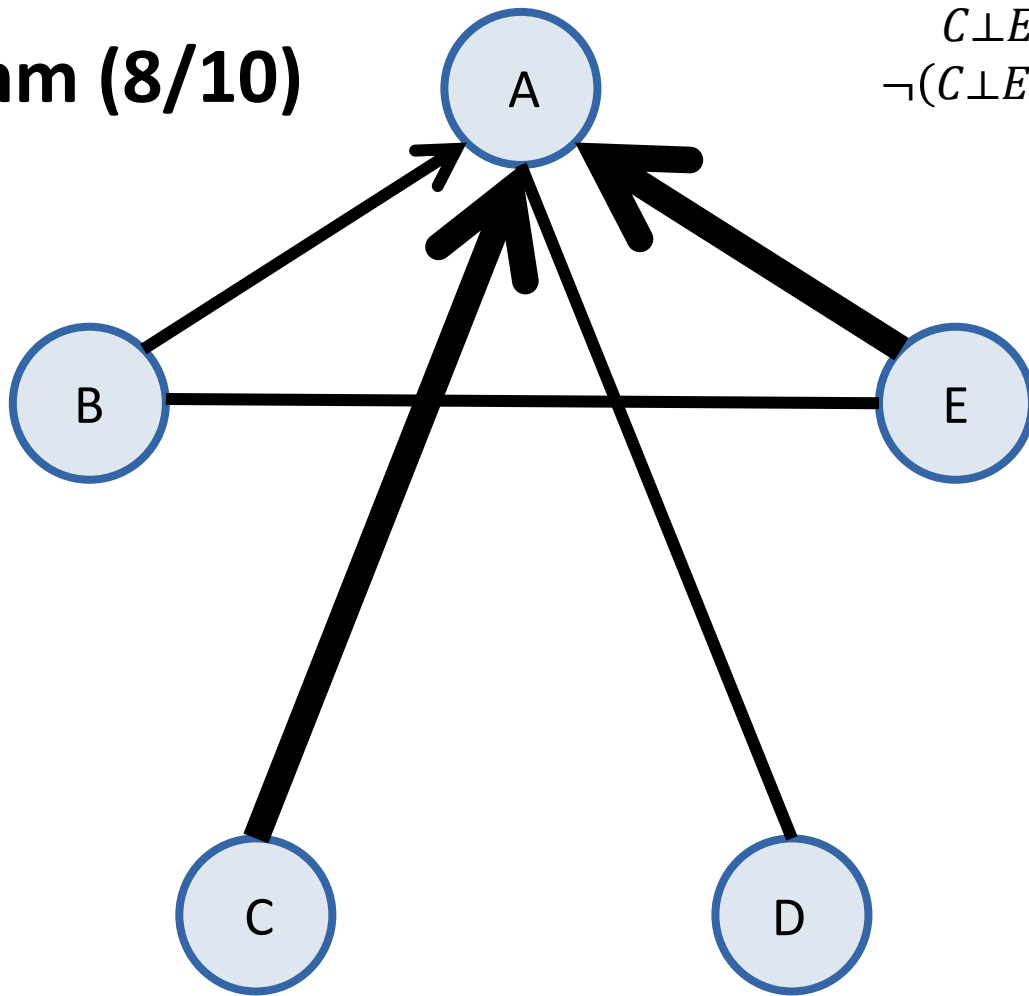


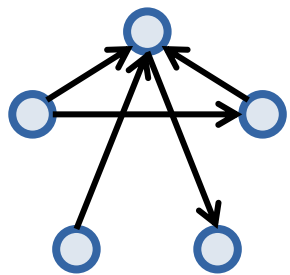
PC Algorithm (8/10)

$$C \perp E$$

$$\neg(C \perp E) | A$$

- Start with a complete (undirected) graph
- Remove the edge $X - Y$ if $X \perp Y | Z$ for some (possibly empty) set of nodes Z
- For all $X - Z - Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y) | Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders

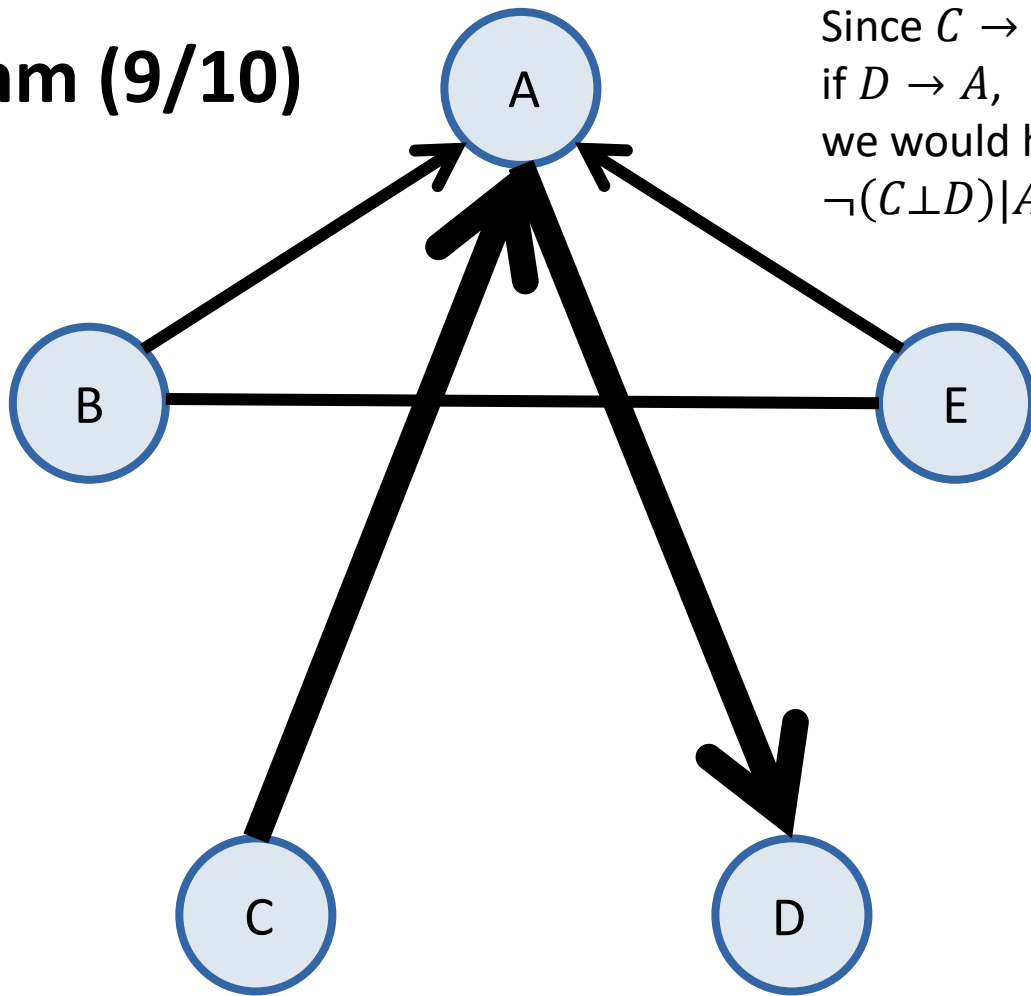


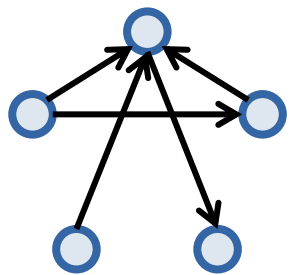


PC Algorithm (9/10)

Since $C \rightarrow A$,
if $D \rightarrow A$,
we would have
 $\neg(C \perp D) | A$

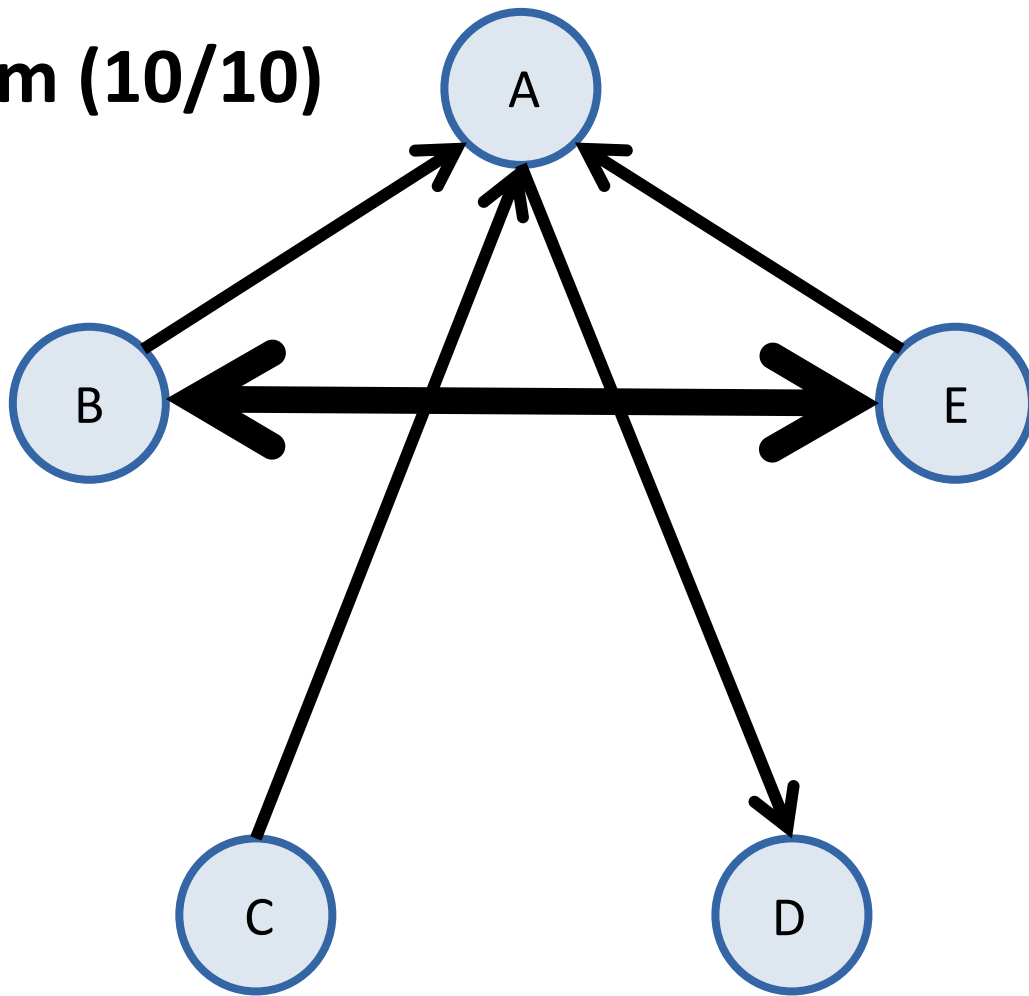
- Start with a complete (undirected) graph
- Remove the edge $X - Y$ if $X \perp Y | Z$ for some (possibly empty) set of nodes Z
- For all $X - Z - Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y) | Z$, we have a collider
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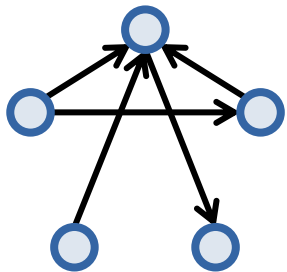


PC Algorithm (10/10)

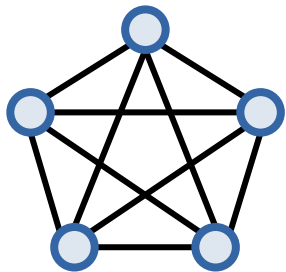
- Start with a complete (undirected) graph
- Remove the edge $X—Y$ if $X \perp Y|Z$ for some (possibly empty) set of nodes Z
- For all $X—Z—Y$ (with no edge between X and Y), if $X \perp Y$ and $\neg(X \perp Y)|Z$, we have a collider
- Propagate the orientation, assuming we have found all the colliders



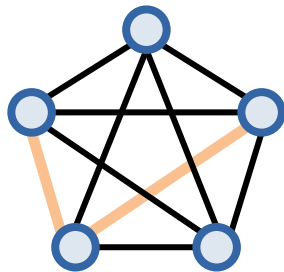
PC Algorithm At A Glance



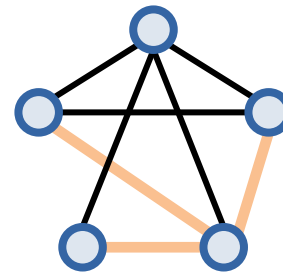
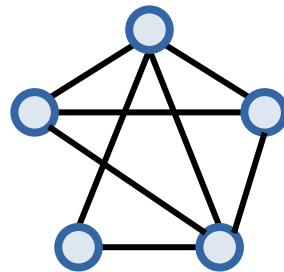
Ground truth



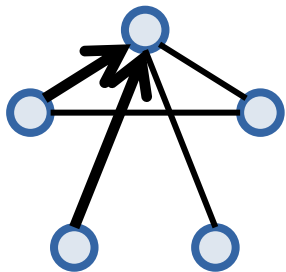
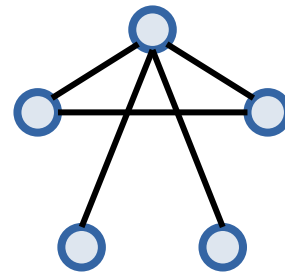
Complete graph



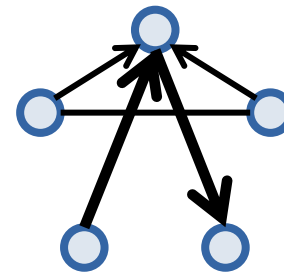
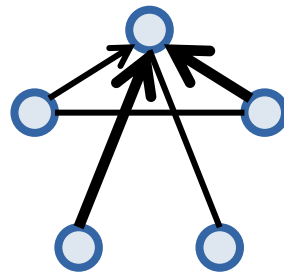
Independence tests



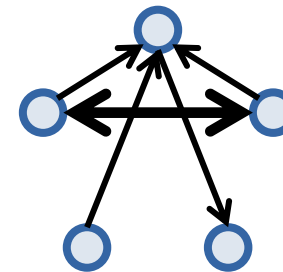
Conditional independence tests



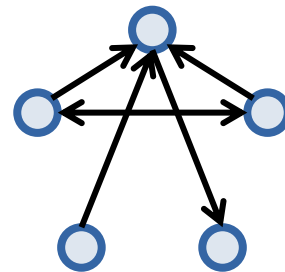
Colliders



Non-collider



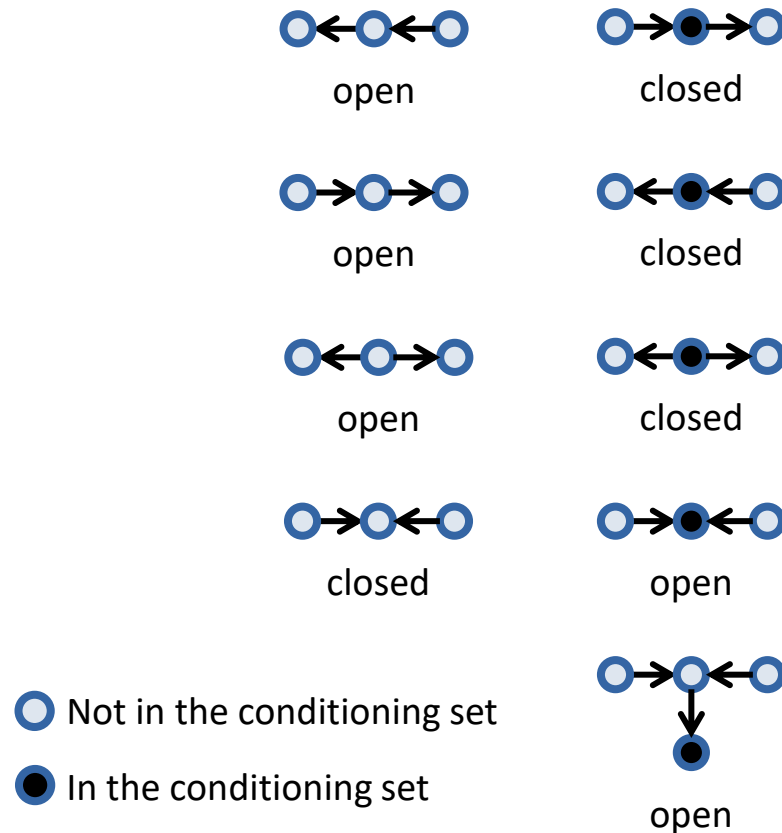
Undirected edge



Final result

Specification Selection (1/2)

- **The sufficient conditioning set is not necessarily unique**
 - For instance, the parents of X may form a sufficient conditioning set for $X \rightarrow Y$, but it may be needlessly large
- More generally, a sufficient conditioning set is a set of nodes blocking all the non-causal paths from X to Y , while leaving all the causal paths open
- If some variables are not observed, things get more complicated, but do-calculus can tell us if the effect can be estimated from observational data alone, and how

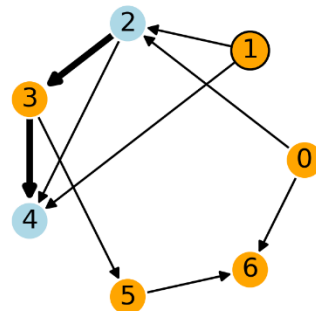


Specification Selection (2/2)

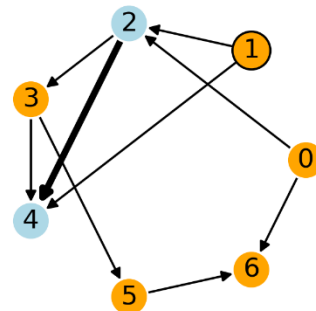
To assess the strength of the causal relation

$X \rightarrow Y$:

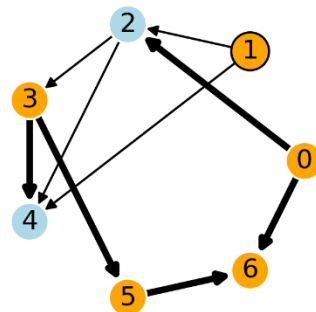
- List all the (undirected) paths from X to Y
- All the non-causal paths should be blocked
 - if not, condition on one or more nodes to block them
- All the causal paths should be open
 - if not, adjust the conditioning set to unblock them



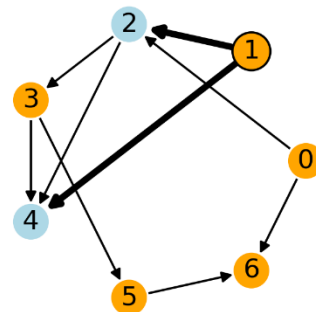
Causal path



Causal path



Blocked by the
collider 6



Blocked if we
condition on 1

Conclusions

Ranking Empirical Evidence

- A scientific theory is a falsifiable statement of the form “ X causes Y through mechanism M ”
- Not all empirical evidence is equal:
 - **Associational evidence** does not suffice to formulate a scientific theory
 - **Causality allows** counter-factual reasoning
- **The adoption of causal inference will help identify the optimal interventions to mitigate climate change**

Type	Rigor	Example
Randomized controlled trials	Very high	Randomly assigning plots to receive different levels of fertilizer or irrigation to measure greenhouse-gas emissions from soil
Natural experiments	High	Comparing regions before and after a sudden policy change (e.g., carbon tax) to estimate its impact on emissions or air quality
Simulated interventions	Medium	Using causal models to simulate the effect of doubling renewable capacity or restoring forests on global temperature trends
Associational studies	Low	Observing that higher CO ₂ concentrations coincide with higher temperatures, without accounting for other drivers such as solar variability
Case studies	Very low	Examining a single glacier's retreat and attributing it to climate change without comparative data
Expert opinion	Anecdotal	Scientists asserting that “urbanization causes local warming” based only on experience, not empirical tests

For More Information



Download for free

Causal Factor Investing

(Cambridge University Press, 2023)

Available at:

[https://www.cambridge.org/core/elements/
causal-factor-investing/
9AFE270D7099B787B8FD4F4CBADE0C6E](https://www.cambridge.org/core/elements/causal-factor-investing/9AFE270D7099B787B8FD4F4CBADE0C6E)

About ADIA Lab

ADIA Lab

- In 2022, ADIA established ADIA Lab as an independent research institute
- Through cutting-edge research in Data Science, AI, Machine Learning and High-Performance Computing, ADIA Lab contributes to solving societally important problems



ADIA Lab Research Verticals & Key Projects (1/2)



Climate Science

1. Techno-economic Assessment of Unconventional Water Sources
2. Using AI and Large Language Models (LLM) for explaining and demonstrating climate simulations
3. Promoting interoperability and interdisciplinary in Climate Science: improve access to data, enhance communication between systems, and foster inter-disciplinary collaboration



Health Sciences

1. Quantifying Economic Value for Health
2. Federated Data Tools for Secure Health Data Sharing
3. Redesign Healthcare Delivery
4. Adoption of AI Technologies by Health Professionals
5. Causal AI for the Generation of Real-World Evidence
6. Impact of Diversity on deployment of precision medicine in the UAE



Materials Science

1. AI-Guided Discovery of Advanced Materials for Extreme Environments
2. Accelerated Simulation Platforms for Materials Design and Testing
3. Development of Sustainable and Recyclable Materials for Industrial Use
4. Integration of Quantum Computing for Predictive Materials Modeling
5. Autonomous Laboratory Systems for Rapid Materials Innovation

ADIA Lab Research Verticals & Key Projects (2/2)



Digital Economy

1. Trusted digital contract framework for the digital economy
2. Labor and Capital Mobility in the Production Network
3. Phygital Asset Integration Framework: Creating and Securing Digital Representations of Physical Assets



Trustworthy AI

1. Robust and Transparent AI Models for High-Stakes Decision-Making
2. Bias Mitigation and Fairness Evaluation in Machine Learning Systems
3. Causality, Explainability and Interpretability in AI Models
4. Adversarial Robustness and Security in AI Pipelines
5. Certification and Auditing Frameworks for AI Systems



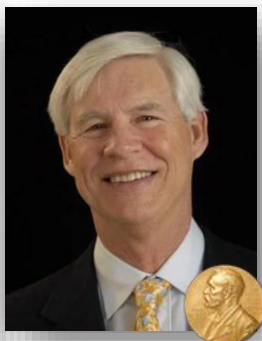
Supercomputing

1. Development of Exascale Computing Capabilities for Scientific Discovery
2. Energy-Efficient Algorithms for Large-Scale Simulation and Modeling
3. AI-Accelerated High-Performance Computing Workflows
4. Advanced Architectures for Quantum-Classical Hybrid Systems
5. Benchmarking and Optimization of HPC Infrastructure

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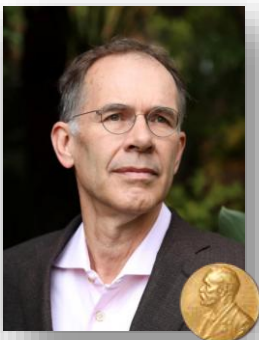
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